Yona BELAREDJ-LAPEYRE
Curriculum Vitae

Centre de Recherche Astrophysique de Lyon
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http://y-lapeyre.github.io/

RESEARCH EXPERIENCE

2024 - present Centre de Recherche Astrophysique de Lyon (CRAL), France
"Simulating the evolution of warped black hole accretion discs."

2026 Theoretical Astrophysics Laboratory, Nagoya University
4 month research visit in Prof. Inutsuka's group.
Developed a Godunov Smoothed Particle Hydrodynamics Solver (Godunov SPH) and started a fixed-metric general relativity SPH (GRPH) solver for multi-GPU architectures.

2022 Kavli Institute for the Mathematics and Physics of the Universe, Tokyo University
4-months research internship in the Center for Data Driven Discovery group.
"How does the intrinsic alignment of galaxies evolve as galaxies age?"
Supervisors: Jinijing Shi, Jia Liu

EDUCATION

2024 - present Ph.D in Astrophysics
Centre de Recherche Astrophysique de Lyon (CRAL), France
PhD candidate in the AstroENS group.
Supervisor: Guillaume Laibe
Expected graduation: August 2027

2020-2024 Bsc & Msc of Theoretical Physics
Ecole Normale Supérieure de Lyon (ENS), France
Top French University for research training.

PUBLICATIONS

Published
The SHAMROCK code: I – smoothed particle hydrodynamics on GPUs
T. David-Cléris, G. Laibe, ***Y. Lapeyre***, MNRAS, https://doi.org/10.1093/mnras/staf444

In preparation
Resolved simulations of teared black hole accretion disks
Y. B. Lapeyre, *G. Laibe, T. David—CLéris, R. Nealon, D. Price*

A multi-GPU Godunov Smooth Particle Hydrodynamics solver in Shamrock

Y. Guo, ***Y. B. Lapeyre****, SI. Inutsuka*

Smoothed Particles Magnetohydrodynamics on the GPU
Y. B. Lapeyre, *G. Laibe, D. Price*

CONFERENCES & SEMINARS

[Seminar] Astro seminar, Kavli IPMU, University of Tokyo, July 2026

[Talk] UK & Ireland Disk meeting, DAMTP, University of Cambridge, July 2026

[Talk] SF2A days 2026, IPAG, Grenoble, June 2026

[Seminar] Astro seminar, TA-Lab, July 2026

[Seminar] Dynamical Lunch seminar, IoA, University of Cambridge, November 2025

[Talk] 2nd European Phantom code family users workshop, IPAG, Grenoble, June 2025

[Poster] International Conference on Exoplanets and Planet Formation, Tsung-Dao Lee Institute, December 2025

[Poster] SF2A, Toulouse, July 2025

[Poster] Advanced summer school on Mathematical Fluid Dynamics, Cargèse, April 2025

TEACHING

2024-2026 Université Claude Bernard Lyon 1
Teaching Assistant. Total teaching time:
- 2nd year level: *Programmation in C++*
- Master 1 Level: *Numerical modelling in Astrophysics/Cosmology*, *Fundamentals of Mathematics for Geophysics*

TIME ALLOCATION

2025: CoI of a GENCI project: 250k GPU hours on the Adastra partition (PI: G. Laibe)
Ran Bardeen-Petterson simulations (B.Lapeyre et al, in prep)

2025: PI of an Argonne project (dynamical allocation): 20k GPU node hours on Aurora
Developed the accretion disc physics and the ideal MHD solver

REFERENCES

Dr. Guillaume Laibe (PhD supervisor), ENS de Lyon, France guillaume.laibe@ens-lyon.fr
Dr. Daniel Price, Monash University, Australia daniel.price@monash.edu
Dr. Shu-Ichiro Inustuka, Nagoya University, Japan inutsuka@nagoya-u.jp